CS-104 Quiz 3

Fall 2026

Name: _____

This is an in-class, closed-book quiz. If you have any questions, please feel free to ask them. When you are finished, you can turn it in and go.

1. (8 points)

Write a function called `trapezoid_area` that receives the lengths of the two parallel sides (`base1` and `base2`) and the height (`height`) of a trapezoid, and returns its area. This can be computed as:

area = average_base × height

where `average_base` is the average (mean) of `base1` and `base2`. Be sure to use appropriate Python syntax. (You do not need to provide an algorithm, documentation, or test cases.)

2. (4 points)

Assume a function `double(x)` has already been defined that returns twice the value of `x`. Write a code segment that uses (calls) this function to compute and print the sum of `double(3)`, `double(5)`, and `double(10)`.

3. (4 points)

Consider the following code:

```
import numpy as np
values = [1, 2, 3]
arr = np.array(values)
print(values * 2)
print(arr * 2)
```

What does each `print` statement output, and why are the two results different?

4. (4 points)

The following code is supposed to plot a labeled graph of $y = x^2$, but running it produces a plot with no axis labels or title. Add the line(s) needed to give the plot an x-axis label of `"x"`, a y-axis label of `"y"`, and a title of `"Squares"`.

```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(0, 10, 100)
y = x ** 2

plt.plot(x, y)
plt.show()
```